Telecom Customer Segmentation and Churn Predictive Analysis

Customer Profiling, Churn Prediction and Retention Strategies

Executive Summary

Nearly half of customers discontinued their subscriptions in their first year of service, while customers paying more than $70 per month had a 35.15% churn rate. These findings highlight two critical retention challenges for telecommunications providers: retaining newly acquired customers and demonstrating value to high-spending subscribers. This analysis examined 7,043 customer records using statistical hypothesis testing, K-Means clustering, and Artificial Neural Network (ANN) modeling to identify churn patterns, segment customers, and predict churn risk.

The analysis identified four distinct customer segments, with new independent customers recording the highest observed segment churn rate at 51.0%. The ANN model achieved 76.0% churn recall and a ROC-AUC of 0.75 on an independent hold-out dataset, showing potential to identify customers at risk of cancellation. The findings support three targeted retention priorities: strengthening early customer engagement, improving perceived value among high-spending subscribers, and encouraging high-risk customers to adopt long-term contracts. Integrating predictive insights with measurable retention initiatives can help the business make more informed decisions to reduce customer attrition, protect recurring revenue, and strengthen long-term customer relationships.

Customer Segments

K-Means clustering placed our customer population in four segments. The cluster labels below describe the dimensions first and then the business persona, avoiding the assumption that the names themselves are model outputs.

Table 1

Customer Segments Characteristics

Customer Segment	Characteristics	Observed churn	Business priority
Cluster 0: New Independent Customers	100% no dependents; 100% short tenure; monthly charges mixed	51.0%	Early onboarding and rapid service recovery
Cluster 3: Mid-Tenure Independent Customers	100% no dependents; 100% mid-term tenure; majority high charges	27.3%	Renewal, loyalty and value perception
Cluster 1: Family Value Customers	100% with dependents; all tenure bands; majority low-to-medium charges	17.1%	Protect retention and improve household value
Cluster 2: Loyal Premium Customers	100% long tenure; 74.7% no dependents; 74.3% high charges	11.1%	Protect loyalty and encourage advocacy

These are observed cluster churn rates, not causal effects. Cluster 0 is the immediate retention priority. Cluster 2 combines the lowest observed churn with a high concentration of premium spending, while Cluster 1 is comparatively stable and more value-oriented.

Predictive Analysis

The dataset exhibited a class imbalance, with 73.18% of customers remaining active and 26.82% discontinuing their subscriptions. This imbalance posed a modeling challenge because a classifier trained on the original distribution favored the majority class and failed to identify a substantial proportion of customers who ultimately churned. To address this limitation, we applied the Synthetic Minority Over-sampling Technique (SMOTE) to the training data to improve the model's ability to learn patterns associated with the minority churn class.

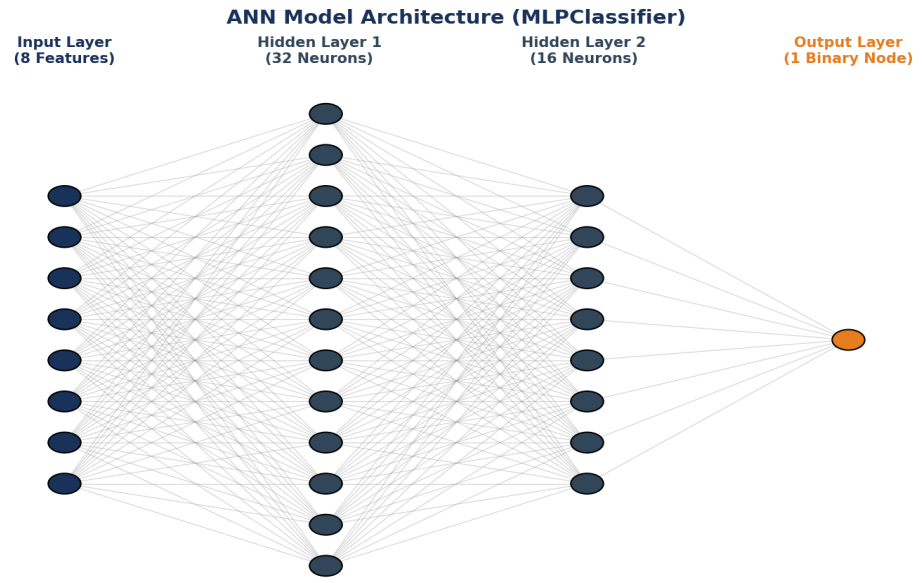
Exploratory analysis showed a strong association between dependents, tenure, monthly charges, and churn. Therefore, our predictive model used one-hot-encoded versions of dependents, tenure bands, and monthly charges bands as ANN inputs. The project reserved 10% of records (705 customers) for a production-simulation holdout and divided the remaining 90% into an 80/20 training/test split (5,070 and 1,268 customers, respectively). Additionally, we built data preprocessing and model pipelines to perform feature engineering and predict customer churn on the hold-out data as if in a production environment.

ANN Architecture

We developed and evaluated a Multi-Layer Perceptron (MLP) classifier, emphasizing recall for the churn class. The classifier has 8 encoded input features and two hidden layers with 32 and 16 neurons. We adopted the following training parameters: ReLU activation, Adam optimization, max_iter=300, random_state=42, early_stopping=True, and validation_fraction=0.1.

Figure 1

ANN Architecture

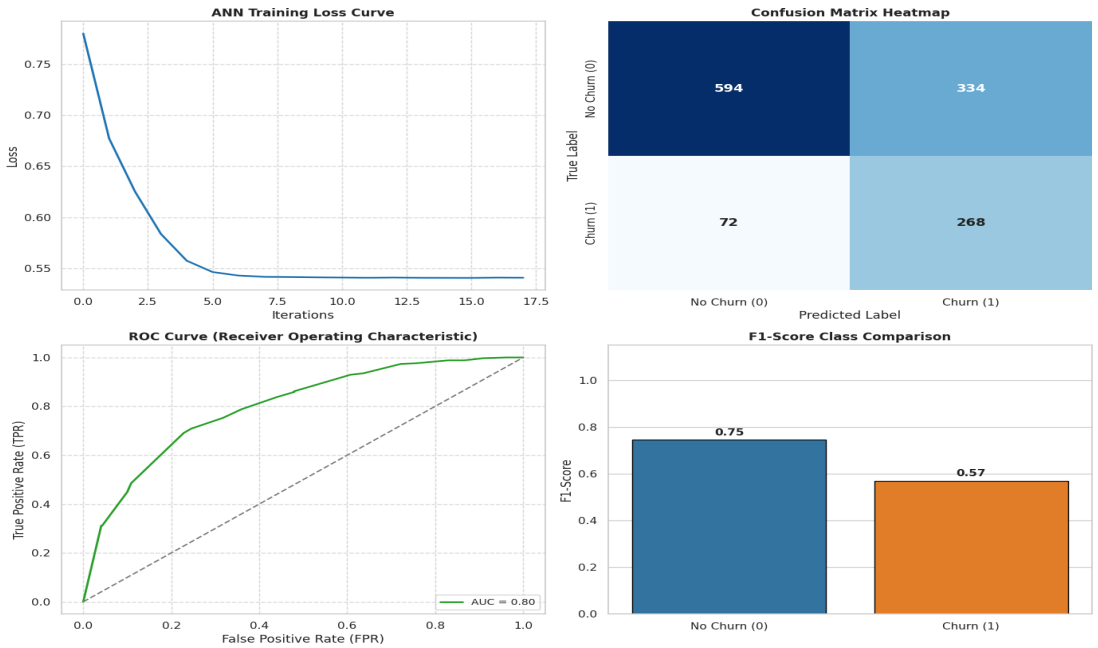


The model achieved overall accuracies of 67.98% on the validation dataset and 64.82% on the hold-out dataset. Although these figures indicate moderate classification performance, this report interprets them in the context of the model's emphasis on identifying churners rather than maximizing overall accuracy. The precision for the churn class ranged from 38.0% to 45.0%, suggesting that a considerable proportion of customers flagged as high risk may not ultimately churn. This trade-off is potentially acceptable in a retention program where the cost of a targeted intervention is lower than the financial impact of losing a customer. However, campaign costs and retention outcomes should be measured to confirm that the approach generates positive returns.

The model also achieved an Area Under the Receiver Operating Characteristic Curve (ROC-AUC) of 0.75, indicating a reasonable ability to distinguish between customers who churn and those who remain active. Overall, the results demonstrate that the ANN model can support churn risk identification and customer prioritization, although its predictions should complement, rather than replace, business judgment.

Figure 2

Model Evaluation Metrics

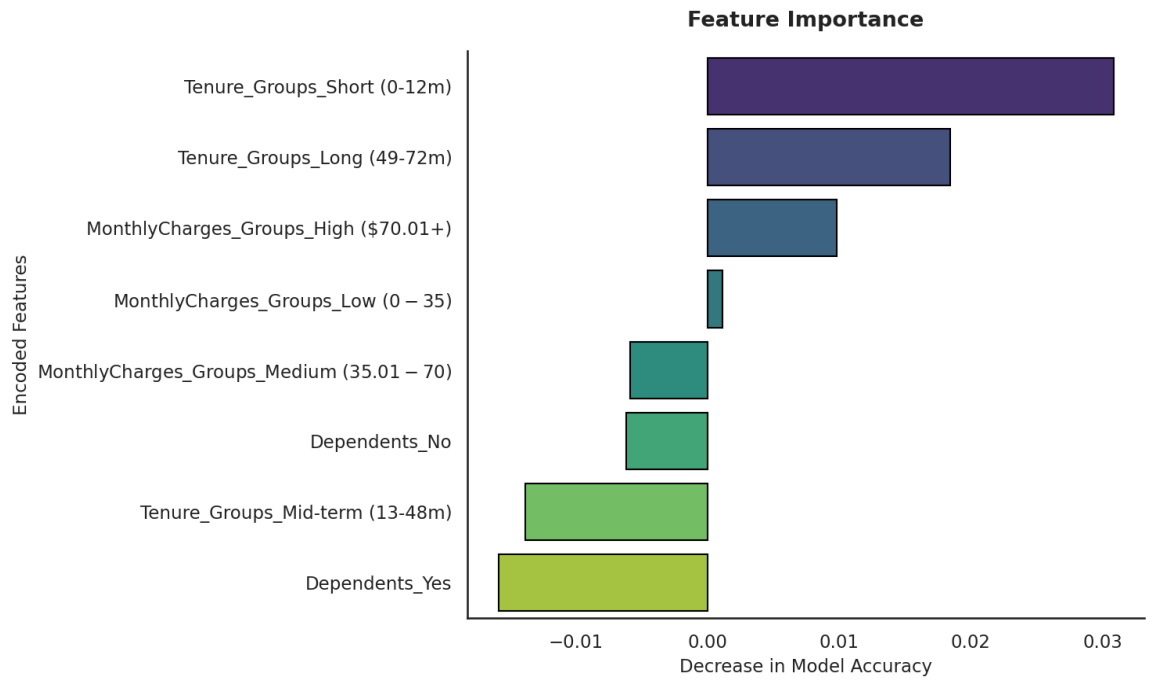


Factors Driving Customer Churn and Retention

The analysis identified customer tenure, monthly charges, and the presence of dependents as important characteristics associated with churn. These findings provide a basis for understanding differences in customer behavior and developing targeted retention strategies. As shown in Figure 3, short tenure was the strongest predictor in the ANN model, followed by long tenure. Monthly charges and dependents also contributed to the model, although their importance was comparatively lower.

Figure 3

Feature Importance



Customer Tenure

The first year of the customer relationship emerged as a critical retention period. Customers with 0–12 months of service recorded a churn rate of 47.44%, indicating substantial vulnerability among newly acquired subscribers.

This pattern may reflect onboarding difficulties, unmet service expectations, or limited perceived value. The business should strengthen early customer engagement, monitor satisfaction, and resolve service issues before they lead to cancellation.

High Monthly Charges

Customers paying more than $70 per month recorded a churn rate of 35.15%, compared with 11.26% among customers paying 0–35. This difference highlights high-spending customers as an important group for targeted retention efforts.

Higher monthly expenditure may increase sensitivity to price and perceived value, particularly when customers feel that service benefits do not justify the cost. The business should conduct proactive account reviews to assess package suitability, service utilization, and opportunities to restructure subscriptions while preserving customer value and profitability.

Customer Dependents

Customers without dependents were more likely to churn than those with dependents. This pattern suggests that household circumstances may be associated with differences in subscription stability and service needs. Customers with dependents may rely on telecommunications services for multiple household members, increasing the practical value of maintaining their subscriptions.

However, dependents should not be interpreted as a direct cause of retention. The relationship may interact with tenure, pricing, service requirements, and contract type. The business should therefore use this characteristic alongside other customer information when prioritizing retention interventions.

Low Monthly Charges

Customers paying between $0 and $35 per month had a churn rate of 11.26%, or a retention rate of 88.74%. This makes the low-cost category comparatively stable within the analyzed population.

Lower expenditure may improve affordability and reduce the financial burden of maintaining a subscription. The business should preserve this segment's value proposition while identifying opportunities for relevant service upgrades. Such offers should reflect demonstrated customer needs and provide clear additional value rather than simply increasing monthly expenditure.

Taken together, the findings suggest that tenure, monthly charges, dependents, and contract arrangements can help differentiate higher-risk customers from more stable segments. These characteristics should inform retention decisions alongside the ANN model's predicted churn probabilities.

Recommendations

The findings support three targeted retention initiatives: early customer engagement, value optimization for high-spending customers, and migration to longer-term contracts. The ANN model can help prioritize customers for these interventions, enabling the business to allocate retention resources according to predicted risk and commercial value.

Early Customer Engagement

Establish a structured onboarding and engagement program for customers in their first 12 months of service, prioritizing those flagged as high risk by the ANN model. Given the 47.44% churn rate among short-tenure customers, early intervention should be central to the retention strategy.

Schedule proactive customer interactions around days 30 and 90 to assess satisfaction, identify onboarding difficulties, clarify package benefits, and resolve service concerns before

they lead to cancellation. Where commercially justified, introduce targeted loyalty incentives, such as temporary discounts on relevant add-on services, to strengthen engagement and perceived value.

Evaluate the program using early-tenure churn rates, customer engagement, offer acceptance, and retention outcomes against a suitable comparison group. This will help determine whether the intervention reduces attrition sufficiently to justify its cost.

Optimize Value for High-Spenders

Implement a proactive value optimization campaign targeting customers paying more than $70 per month who are also classified as high risk by the ANN model. With an observed churn rate of 35.15%, this group warrants focused attention to keep subscription costs aligned with customer needs and perceived benefits.

Conduct personalized account reviews to assess service utilization, package suitability, and opportunities to restructure bundles or move customers to more appropriate service tiers. Where a lower-cost package better meets a customer's needs, offer it as a retention option. Bundling eligible services may also improve perceived value when the additional services provide genuine benefits.

Measure campaign effectiveness through retention rates, changes in average revenue per user, customer satisfaction, and contribution margin. This approach will help establish whether improved retention justifies the cost of incentives or any reduction in subscription revenue.

Contract Migration and Loyalty Offers

Develop a contract migration program for high-risk customers on month-to-month subscriptions. Prioritize customers using ANN-predicted churn probabilities and offer eligible subscribers the option to move to one-year or two-year contracts in exchange for a clearly defined benefit, such as a one-month service credit or another appropriate loyalty incentive.

Before implementing the campaign at scale, run a controlled pilot comparing customers who receive the offer with a comparable group that does not. This will help determine whether contract migration improves retention and whether the associated incentives are commercially sustainable.

Assess the results using contract uptake, churn rates, retention outcomes, and the financial impact of incentives. Expand the program only if the pilot demonstrates meaningful retention improvements and a sustainable business case.

Limitations

Class Imbalance in the Dataset

A principal modeling challenge was the imbalance between active and churned customers. Active customers represented 73.18% of the dataset, compared with 26.82% who churned. This distribution could cause the classifier to favor the majority class, reducing its ability to identify customers at risk of cancellation.

To address this issue, SMOTE was applied to the training data to generate synthetic examples of the minority churn class. This helped the MLP classifier learn churn-related patterns and supported the model's emphasis on recall. The resulting churn recall of 79.0% on validation data and 76.0% on the independent hold-out dataset indicates that the model identified a substantial proportion of churn cases.

Limited Predictive Features

The model relied on a relatively small set of customer characteristics that showed statistically significant association with churn, including dependents, tenure, and monthly charges. Although these variables provided useful predictive information, they may not capture the full range of behavioral and operational factors associated with customer churn.

Future model development should incorporate additional variables when available, reliable, and appropriate for the intended use. Potential enhancements include payment delays, customer complaints, technical support requests, service interruptions, usage patterns, and changes in customer consumption.

These additional features could better represent customer behavior and improve the model's ability to distinguish between customers likely to remain active and those who may cancel.

Classification Threshold Selection

The model's classification threshold determines the probability at which a customer is classified as being at risk of churn. A default threshold of 0.5 provides a convenient starting point, but it may not represent the most appropriate operating threshold for a retention program.

Because the model was developed to prioritize recall, lowering the threshold may enable the business to identify more potential churners. However, this will generally increase false positives, potentially raising campaign expenditure and placing additional demands on retention teams.

The appropriate threshold should therefore be selected by considering the relative costs of missed churn cases, unnecessary interventions, and available retention resources. Management should evaluate alternative thresholds against precision and recall.

Conclusion

This analysis demonstrates how customer segmentation and predictive modeling can help telecommunications businesses identify churn risks and implement retention efforts. Short tenure, high monthly charges, and no dependents emerged as key churn factors. Recommended priorities include early customer engagement, value optimization for high-spending subscribers, and targeted contract migration. The next step should be to test these interventions through controlled campaigns, measure their financial impact, and refine the model with additional behavioral data. Success should ultimately be measured not only by improved churn detection but by demonstrable gains in customer retention and profitability.